

Women's Clinic Finder

1.1 Project Aim

This project focused on developing a web application to help users find clinics specialising in women's health. The application enables users to identify both NHS and private providers for specific conditions, supporting quicker access to relevant healthcare information.

During planning, it became clear that many individuals rely on general search engines for healthcare support. However, these platforms often return overwhelming and inconsistent results, posing challenges for vulnerable users, older adults and those with lower digital confidence.

The Women's Clinic Finder brings healthcare information and clinic searching together in a single application, allowing users to quickly identify suitable services based on their location and healthcare needs.

1.2 Project Objectives

The main objective is to create an efficient app that helps users find suitable women's health services by location, healthcare concerns, and provider type.

To achieve this, the project focuses on the following objectives:

- Develop a postcode-based search so users can find clinics within a chosen distance.
- Allow users to filter results by radius (e.g., 5, 10, 25, 50 miles).
- Organise women's health concerns into clear categories for easy selection.
- Include key categories: General Health, Family Planning, Contraception, Sexual Health, Fertility, Pregnancy, and Menopause Support.
- Allow filtering by provider type: NHS, private, or all.
- Display all relevant clinic results based on the user's location, health concern, provider, and radius.
- Provide a simple interface to reduce the need for multi-site searching.
- Present clinic information clearly to support informed decisions.

These objectives keep the focus on making women's healthcare services easier to find, compare, and access.

1.3 Project Overview

This report documents the planning, design, development, and evaluation of the Women's Clinic Finder application. It begins by exploring the project's motivation and the challenges users face when searching for women's healthcare services. The report then discusses the project's requirements, design decisions, and the technologies selected to develop the application. It also explains how the project was implemented as a collaborative effort, highlighting the development process, individual contributions, and challenges encountered. Finally, the report evaluates the application's functionality through testing, reflects on its current limitations, and considers opportunities for future development.

2.1 Project Domain

Despite abundant online information, accessing women's healthcare remains difficult. Conditions such as PCOS, Endometriosis, Menopause, and Fertility issues often present with overlapping symptoms, making diagnosis and long-term management more complex. Specialist care provides personalised support based on an individual's medical history and ongoing healthcare needs.

The Women's Clinic Finder streamlines access to specialist care by combining healthcare information with location-based clinic searching in a single application. Users can learn about different women's health services before locating suitable NHS and private healthcare providers, helping them make informed decisions while simplifying the search process.

2.2 The problem

Although healthcare information is widely available, our team recognised that existing methods of locating women's healthcare services are often fragmented across numerous websites and healthcare providers. Users are frequently required to search independently, compare information from different sources, and determine whether individual clinics offer the specific services they require. General search engines often return large volumes of information without prioritising the most relevant healthcare providers, making it difficult for users to efficiently identify suitable clinics.

This fragmented approach can be particularly challenging for users who are unfamiliar with online healthcare resources or who have recently developed symptoms or received a diagnosis. As a result, finding appropriate healthcare can become unnecessarily time-consuming and difficult.

2.3 The proposed solution

The application enables users to search by location and select a preferred search radius, ensuring results remain relevant to their local area. It then identifies clinics that meet the selected criteria whilst allowing users to filter results by NHS providers, private providers or both. By bringing healthcare information

and clinic search into a single application, the Women's Clinic Finder reduces the time and effort required to find appropriate healthcare support while providing users with reliable, accessible information throughout the search process.

2.4 Technical Requirements

The following functional requirements were identified during the planning stage to ensure the application met its users' needs.

- Allows users to search for clinics using a valid UK postcode.
- Allows users to select a search radius of 5,10,25 or 50 miles.
- Organise women's healthcare services into clearly labelled category cards.
- Filter clinic results by NHS providers, private providers or both.
- Retrieve clinic information based on the selected postcode, search radius, healthcare concern and provider type.
- Display clinic information, including the clinic name, address, contact number and website.
- Validate user input before a search is performed.

2.5 Non-Technical Requirements

The following non-functional requirements were established to guide the development of the application.

- Provide a clean and user-friendly interface
- Ensure the application is responsive across different screen sizes.
- Display clear validation messages when required information is missing
- Organise the application using reusable React components.
- Structure the database using relational tables to reduce duplicate information.
- Design the application so that additional clinics and healthcare categories can be added easily.
- Return search results efficiently through postcode searching and database filtering.

3.1 Architecture and Design Decisions

The Women's Clinic Finder uses a three-tier architecture: a React frontend, an Express.js backend, and a MySQL database. This separation keeps the interface, logic, and data storage independent, making the app easier to maintain and expand.

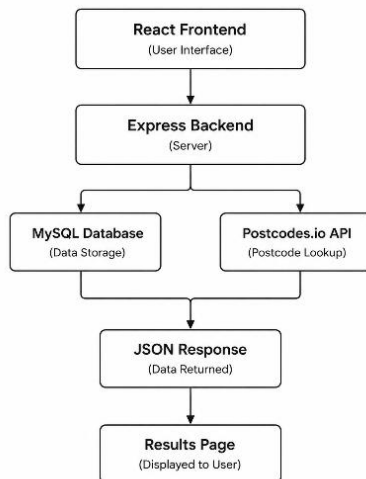


Figure 3.1: System Architecture

As illustrated in Figure 3.1, the frontend captures the user's search criteria before sending the request to the backend. The backend communicates with the Postcodes.io API to retrieve postcode coordinates and queries the MySQL database to return clinics that match the selected location, search radius, healthcare category, and provider type.

The database was designed using three related tables: clinics, health_concerns, and clinic_concerns. The database consists of three related tables: clinics, health_concerns and clinic_concerns. This structure minimises duplicated data and allows clinics to be associated with multiple healthcare categories. This structure minimises duplicated data and allows all clinics to be associated with multiple healthcare categories where appropriate.

Several design decisions were made to improve usability. Category cards were used instead of a traditional drop-down menu to make selecting healthcare concerns more intuitive, while postcode searching with a selectable search radius returns location-based clinic recommendations, and separating the search and results pages presents information in a clear and organised format.

3.2 Technology Choices

Technology	Justification
React	Used to create reusable frontend components, such as the search form and category cards
Express.js	Processes user requests and communicates with the database API.
MySQL	Stores structured clinic and healthcare data using relational tables
Postcodes.io API	Converts UK postcodes to geographic coordinates for location-based search.
Mantine UI	Provides reusable interface components to create a consistent user interface.
CSS Modules	Allows styling to be applied to individual components without affecting the rest of the application.
GitHub	Supports collaborative development through branches and pull requests.
Figma	Used to produce the wireframes during the planning stages before development started.

4.1 Development Approach and Team Responsibilities

The Women's Clinic Finder was developed using an Agile approach, allowing the team to work collaboratively whilst developing the application in incremental stages. GitHub was used for version control, enabling team members to work on separate branches before merging completed features into the main project. Regular communication through Slack, Trello and Zoom meetings ensured that progress was monitored, tasks were allocated and technical issues were resolved efficiently.

Responsibilities were distributed across the team according to each team member's area of the project. Destiny Iyamu Omoragbon developed the application's search functionality, frontend integration and routing. Molly Lester contributed to the project setup, backend development, API integration, user interface design and project documentation. Danielle Brereton-Smith designed and implemented the MySQL database, populated it with clinic data and supported frontend-backend integration. Tia Benvenuti developed the original project concept and produced the wireframes used throughout development. Saamiya Kudah created the project presentation and contributed to styling the results page. Tapiwa Chibagidi was responsible for the project documentation, report writing and developing unit tests using Vitest.

4.2 Tools and Libraries Used

The project was developed using React for the frontend, Express.js and Node.js for the backend, and MySQL for database management. The Postcodes.io API was integrated to support location-based searching, whilst Mantine UI and CSS Modules were used to create a responsive user interface. GitHub supported version control and collaborative development, and Postman was used to test backend API endpoints before frontend integration.

4.3 Key Achievements and Challenges

A key achievement of the project was successfully integrating the frontend, backend, database and external API into a fully functional application capable of returning nearby women's healthcare clinics. Throughout development, the team encountered several challenges, including manually sourcing accurate clinic data, configuring communication between the frontend and backend, resolving database connection and API issues, and learning unfamiliar technologies whilst working collaboratively on a shared codebase.

4.4 Pivots and Decisions

During development, several design decisions were refined as the project progressed. The team initially intended to focus on clinics within underserved areas of the UK; however, this approach was revised due to the limited availability of suitable women's healthcare services. The user interface was also

refined by introducing category cards and provider filters to simplify the search process and improve the overall user experience.

4.5 Agile Practices

The project followed an iterative development process, with features being developed, tested and refined throughout the project. GitHub branches, pull requests, and code reviews supported collaborative development and reduced integration issues. Refactoring was carried out where appropriate to improve code organisation, whilst regular testing ensured that new functionality operated as expected before being merged into the main application.

Testing and evaluation

5.1 Testing Strategy

To verify that the Women's Health Clinic Finder functioned as intended, both manual and automated testing approaches were used throughout development. Manual testing was carried out during each development iteration to ensure that the application's functionality, user interface and API integration behaved as expected. In addition, automated unit testing was carried out using Vitest and React Testing Library to test individual React Components. This helped identify issues early, ensured the application behaved as expected and reduced the likelihood of introducing bugs during development.

5.2 Test Cases

Several unit tests were written to test key functionality within the application. These included confirming that the application and search form rendered correctly, checking that the postcode input field was displayed, and verifying that validation messages appeared when required information was missing. User interaction was also tested by simulating button clicks to ensure the validation worked as expected.

Test Case	Expected Result	Outcome
Application renders successfully	Application loads without errors	Pass
Search form renders	Search form and heading are displayed	Pass
Postcode input renders	User can view postcode input field	Pass
Empty postcode validation	Validation message displayed	Pass
Empty category validation	Validation message displayed	Pass
Search button interaction	Validation triggered when required fields are missing	Pass

5.3 Results

All implemented unit tests passed successfully, confirming that the application's frontend functionality worked as intended. The tests verified that the application rendered correctly, displayed appropriate validation messages and prevented users from searching without completing the required fields. Manual testing of the application's search functionality and API integration also demonstrated that the implemented features worked as intended and met the project's functional requirements.

Women's Health Clinic Finder
Helping you find the right women's health support by filtering local services based on what you need.

Where should we search?
Enter your postcode: 5 miles:

What brings you here today?

General Women's Health GP services, routine check-ups, and general health concerns.	Family Planning & Contraception Support with contraception, pregnancy planning, and reproductive choices.
Sexual Health Testing, treatment, and advice for sexual health and wellbeing.	Fertility Support Help with fertility concerns, testing, and treatment options.
Pregnancy & Maternity Care Support during pregnancy, antenatal care, and postnatal services.	Menopause Support Advice and treatment for menopause symptoms and hormonal changes.

Provider type

☐ All services ☐ NHS ☐ Private

Find My Best Matches

Figure 5.1 Search Interface (left)

Women's Health Clinics Near You

Community Women's Health Service (Barts Health)
(NHS)
Sylvia Pankhurst Centre, Mile End Hospital, Bancroft Road, London, E1 4DG
020 8223 8449
[Visit website](#)

Oxleas Women's and Girls' Health Hub
(NHS)
Market Street Health Centre, Woolwich, SE18 6QR
020 8301 8920
[Visit website](#)

Camberwell Sexual Health Centre
(NHS)
Camberwell Building, 100 Denmark Hill, London, SE5 8RS
020 3299 5091
[Visit website](#)

Figure 5.2 Results Page (right)

5.4 Limitations

Although the implemented unit tests covered key frontend functionality, they focused only on testing the React components. The backend API and database interactions were tested manually rather than through automated tests. End-to-end testing was outside the scope of this project, meaning complete user journeys were not tested automatically. Future development could introduce integration and end-to-end tests to provide broader coverage of the application's functionality.

6. Conclusion

The Women's Clinic Finder successfully met its aim of providing users with a simple and efficient way to find women's healthcare services based on their location, healthcare concern and preferred provider type. By bringing healthcare information and clinic search into a single application, the project reduced the need for users to navigate multiple websites to find appropriate support.

Throughout the project, the team worked collaboratively to design, develop and test the application using an Agile approach. This project provided valuable experience in full-stack development, API integration, database design and collaborative software development, while also strengthening problem-solving and communication skills. Although there is scope for future improvements, including expanding the clinic database and introducing additional automated testing, the application successfully met its objectives and provides a practical solution to a real-world problem.